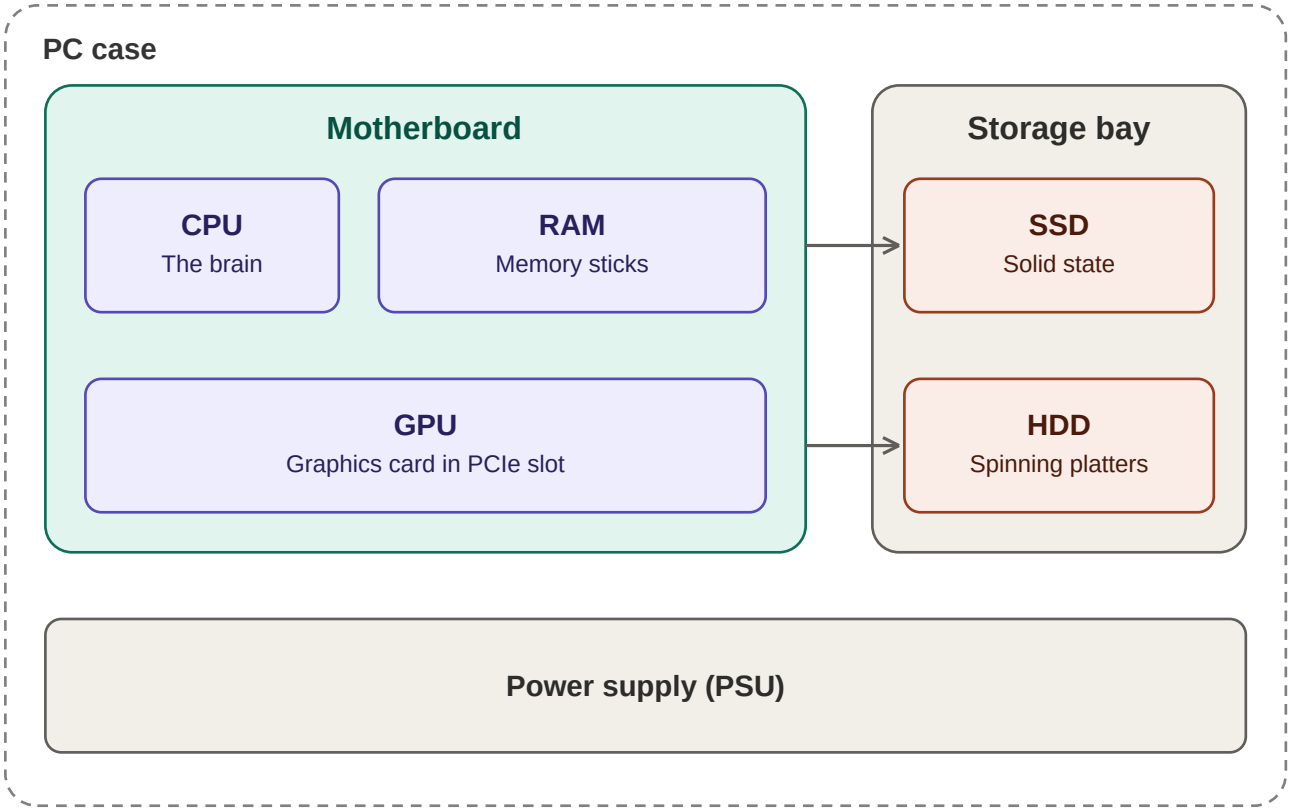
PC Hardware: Tasks and Answers

Task 1: Labeled diagram of a desktop PC

Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



Task 2: HDD vs SSD comparison

Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

	HDD	SSD
Speed	Slower (about 100-200 MB/s); games and Windows load slowly.	Much faster (about 500 MB/s on SATA, 3,000+ MB/s on NVMe).
Durability	Fragile; moving parts can break if dropped or bumped.	More shock-resistant; no moving parts.
Typical use	Cheap bulk storage in desktops, backups, media libraries.	Laptops, gaming PCs, boot drive for the OS and games.

Task 3: Roles of CPU, RAM and GPU in a game like PUBG or Free Fire

Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU: It runs the game logic: player movement, bullet physics, AI, and network data from other players. A weak CPU causes stutters when many players are in the same area.

RAM: It holds the map, textures, and game data the game needs right now, so they load quickly. Too little RAM makes the game lag or crash, especially with other apps open.

GPU: It draws every frame you see: the terrain, buildings, characters, shadows, and effects. A stronger GPU gives higher FPS and better graphics settings.

Task 4: Upgrading a PC for the latest FIFA

Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

GPU (first priority): FIFA is graphics-heavy, with detailed players, stadiums, and lighting. A better GPU gives smoother frame rates at higher settings.

RAM: Go to at least 16 GB, since 8 GB can cause stutters with background apps running.

SSD: Move the game onto an SSD for much faster loading screens and menus. An NVMe drive is best.

CPU: Upgrade only if it is old, since it can bottleneck a new GPU and cause dips during busy matches.

PSU: A new GPU may need more power, so check the wattage to keep the system stable.

Motherboard: Replace it only if the new CPU or RAM is not compatible with your current one.